CURRENT / WIP

Building a Portfolio System

From image ingestion to an adaptive Three.js gallery

TAYLOR PIKE

A local-first publishing editor and expandable virtual gallery

01 / ORIGIN

CURRENT

Why I built it

I wanted my photography portfolio to feel distinct from a conventional image grid or template-based site. After learning Three.js, I began thinking about a virtual gallery where visitors could move through the work rather than only scroll past it.

The gallery also continued an older ambition: since learning to code in high school, I had wanted to build an experimental JavaScript project that connected directly to my creative practice.

> *"I wanted the portfolio to feel uniquely mine and meaningful enough to leave a lasting impression."*

VIRTUAL GALLERY / CONCEPT ORIGIN

A spatial presentation built around movement, scale, and sequence.

02 / EVOLUTION

CURRENT

How the gallery became an editor

What started as a convenience became the operating layer for the portfolio.

The first problem was managing the photographs that would eventually appear inside the gallery. Renaming files, creating multiple sizes, editing paths, and changing JSON by hand would become unmanageable as the library grew.

I began with an import tool and controls for the homepage slideshow and traditional portfolio. Each new part of the website introduced another decision that was better represented through a dedicated interface than direct source-code editing.

01 **Virtual gallery idea**

02 **Image import tool**

03 **Homepage + portfolio curation**

04 **Metadata + crop controls**

05 **About-page composition**

06 **Gallery architecture + wall curation**

07 **Site-wide control panel**

PROJECT LOGIC

Each interface emerged from a real maintenance problem inside the portfolio.

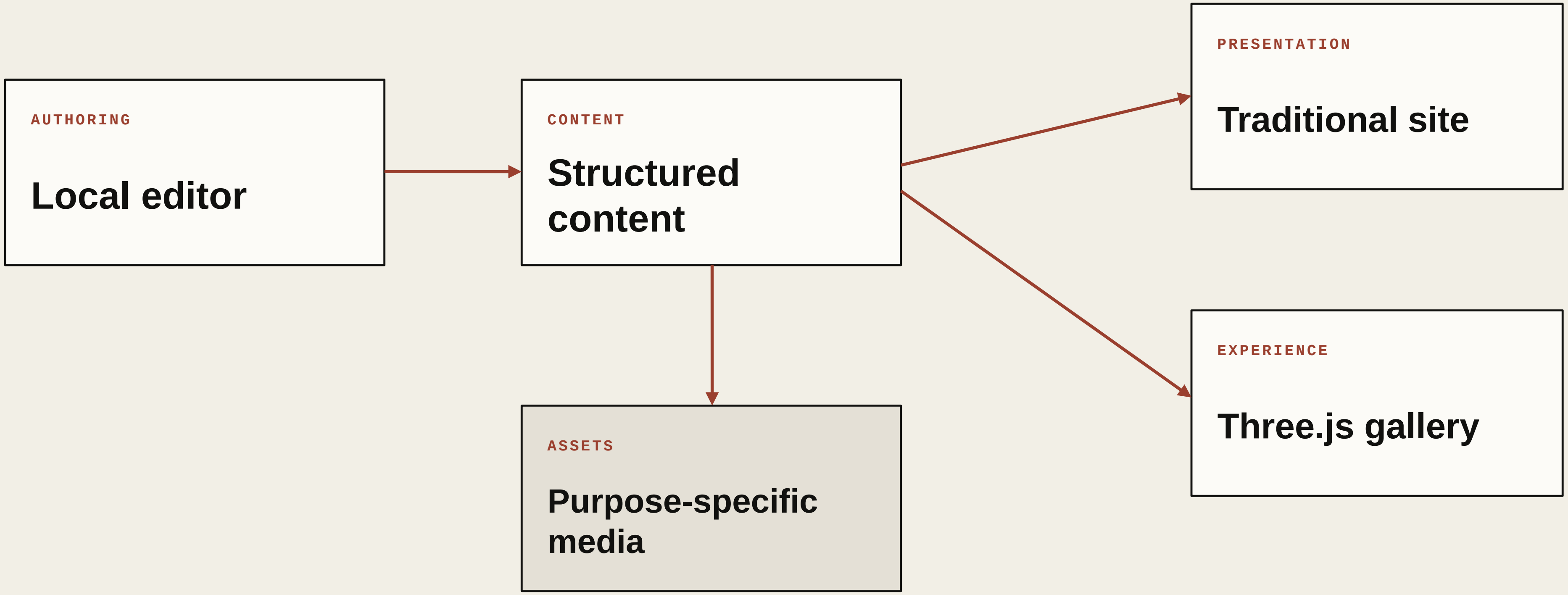
The editor was not designed as a generic CMS first and applied afterward.

03 / SYSTEM

CURRENT

One system, several responsibilities

The portfolio is a connected system: a local editor, structured content, a media pipeline, a traditional public site, and a real-time gallery. Clear boundaries let each part evolve without forcing routine content changes into application code.



SYSTEM RULE

Content can change without rewriting the site.

Specialized interfaces appear only where a conventional form cannot represent the decision clearly.

One content layer supports multiple forms of presentation.

04 / INGESTION

CURRENT

One photograph, four responsibilities

The importer turns a source image into predictable assets and a structured record.

A thumbnail, a gallery texture, a standard display image, and a high-resolution image have different performance and quality requirements. The pipeline creates each rendition for a specific use instead of forcing every page and device to download the same oversized asset.

FAST	BALANCED	BOUNDED	DETAILED
library + grid	public display	Three.js texture	close viewing

INPUT
Source photograph

→

VALIDATE
Review metadata
Category + duplicate checks

→

THUMB	TEXTURE	DISPLAY	FULL
520 px	**1024 px**	**1600 px**	**2400 px**
WebP	WebP	WebP	WebP

→

OUTPUT
JSON record
+ backup
+ source archive

The processed original moves to an imported-source archive after a successful import.

05 / CONTENT

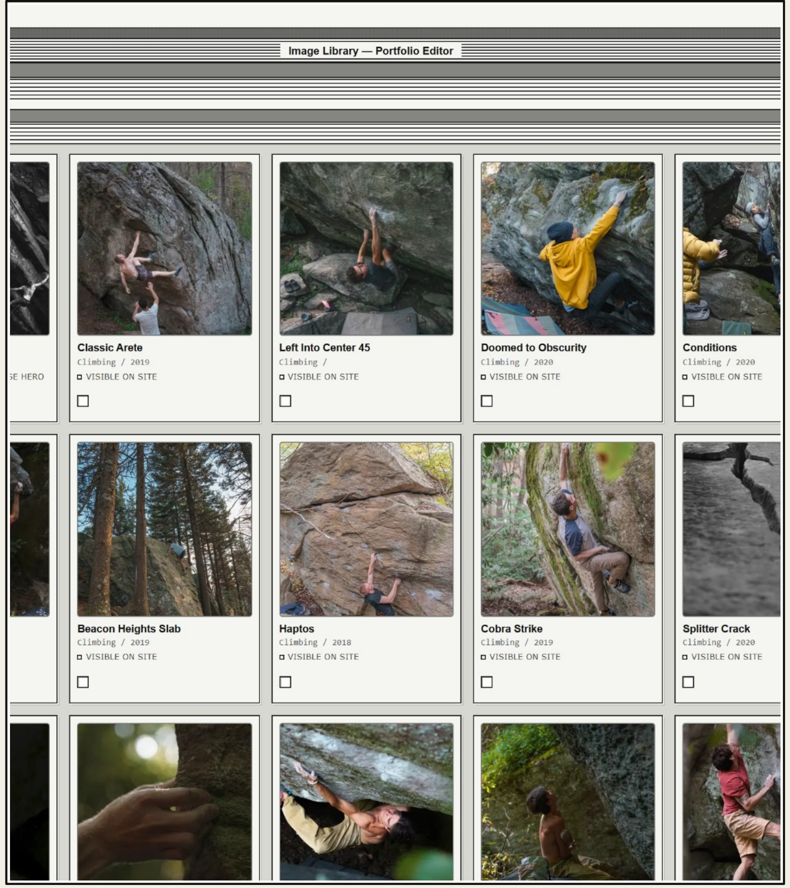
CURRENT

A data-driven website

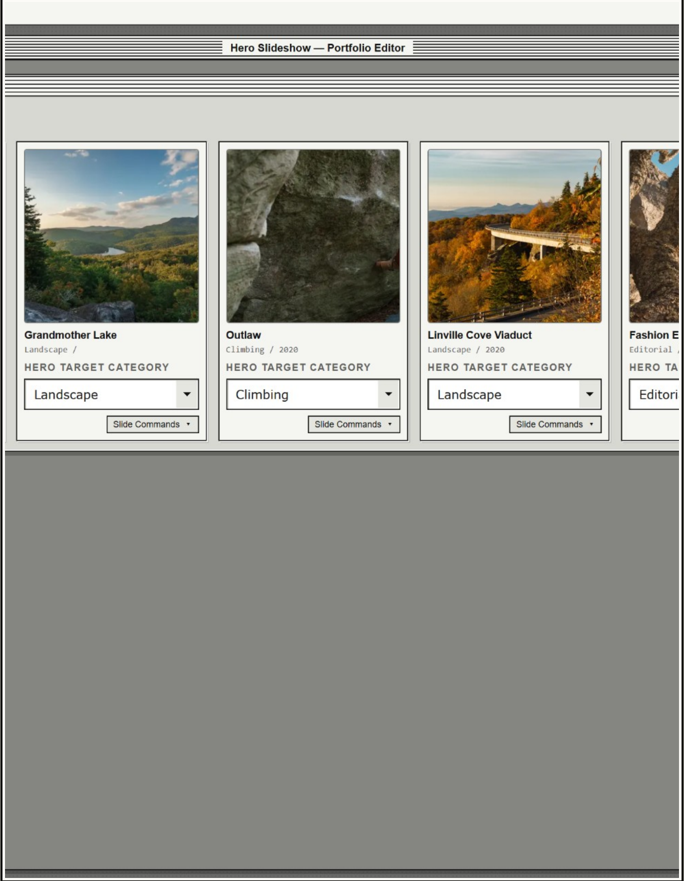
The editor changes the data the website consumes rather than rewriting its visual implementation. That keeps responsive behavior, animation, accessibility, and application logic under developer control while exposing the content that needs regular maintenance.

Images	About photography
Categories	About copy
Hero slides	SEO
Gallery curation	Site copy

IMAGE LIBRARY



HOMEPAGE CURATION



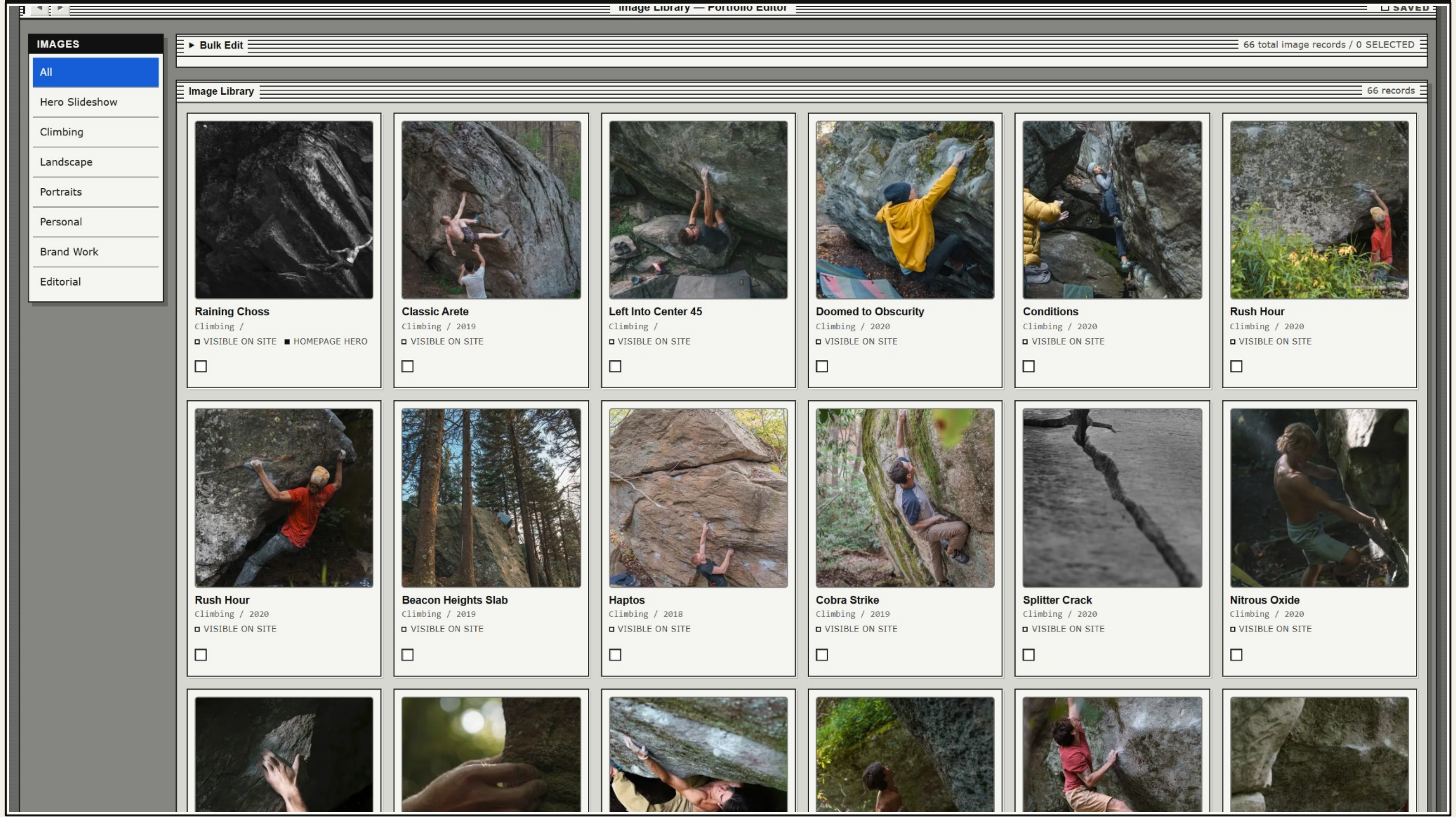
One source of truth can feed multiple presentations.

06 / EDITOR

CURRENT

The editor as a control panel

PORTFOLIO EDITOR / IMAGE LIBRARY



A working tool, not a static product mockup.

CONTROL MODEL

The editor combines structured controls with purpose-built visual workspaces.

Forms handle titles, metadata, categories, copy, labels, and SEO. Visual workspaces handle crops, slideshow ordering, collage composition, and gallery placement.

The result is not unrestricted design software. It is a focused operating layer for one highly customized site.

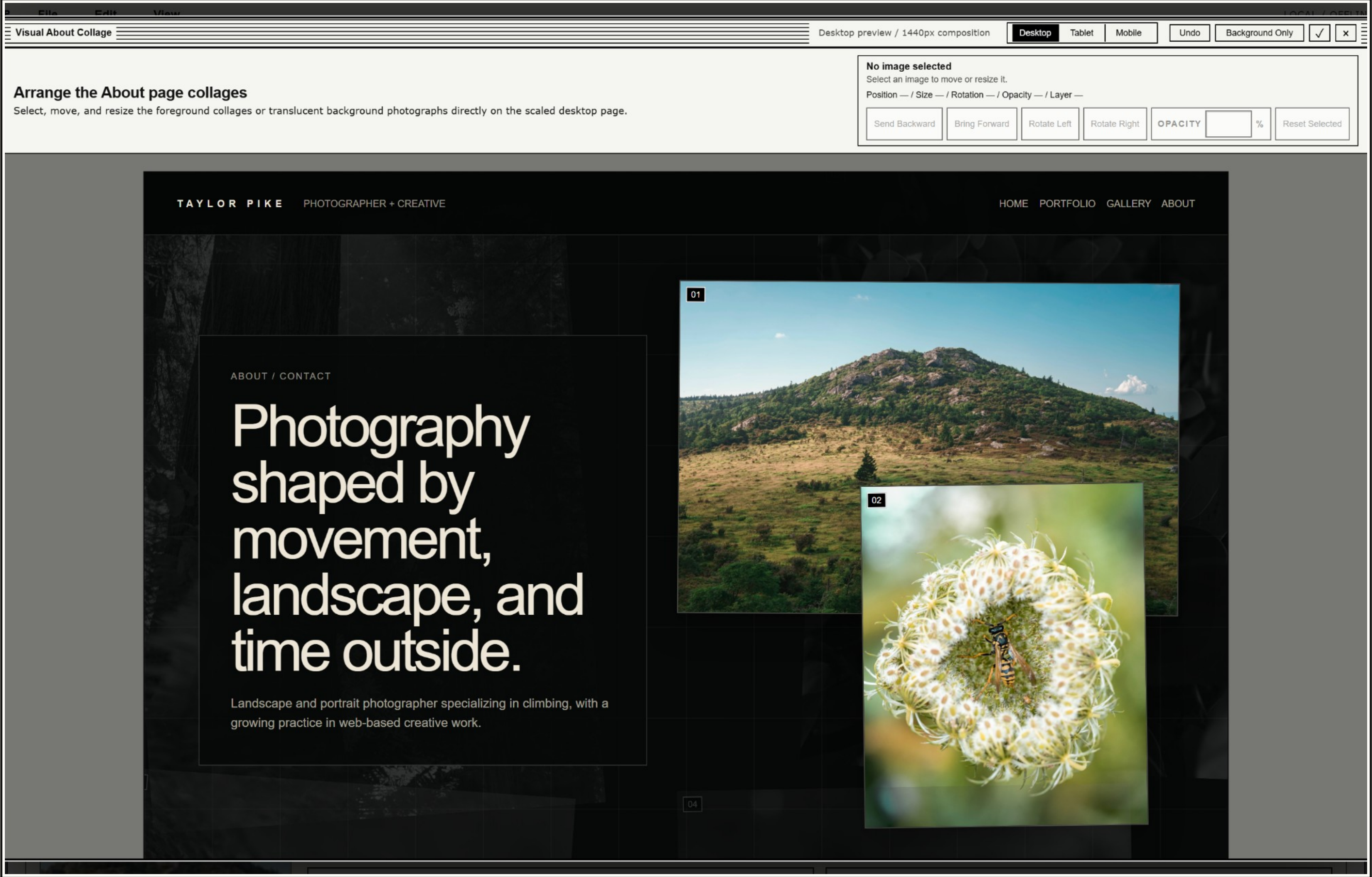
MEDIA	central library
DATA	structured fields
CURATION	site assignments
RECOVERY	save + backups

07 / COMPOSITION

WIP

Bounded visual editing

ABOUT PAGE / VISUAL COMPOSITION



Not every visual decision should become unrestricted design control. The About editor exposes only the controls that matter for this page: position, size, layer, rotation, opacity, and crop.

The user edits presentation data inside a controlled composition. The page structure, responsive logic, accessibility, and larger visual system remain intact.

SAME CONTENT

VIEW 01	VIEW 02
Desktop arrangement	**Mobile arrangement**

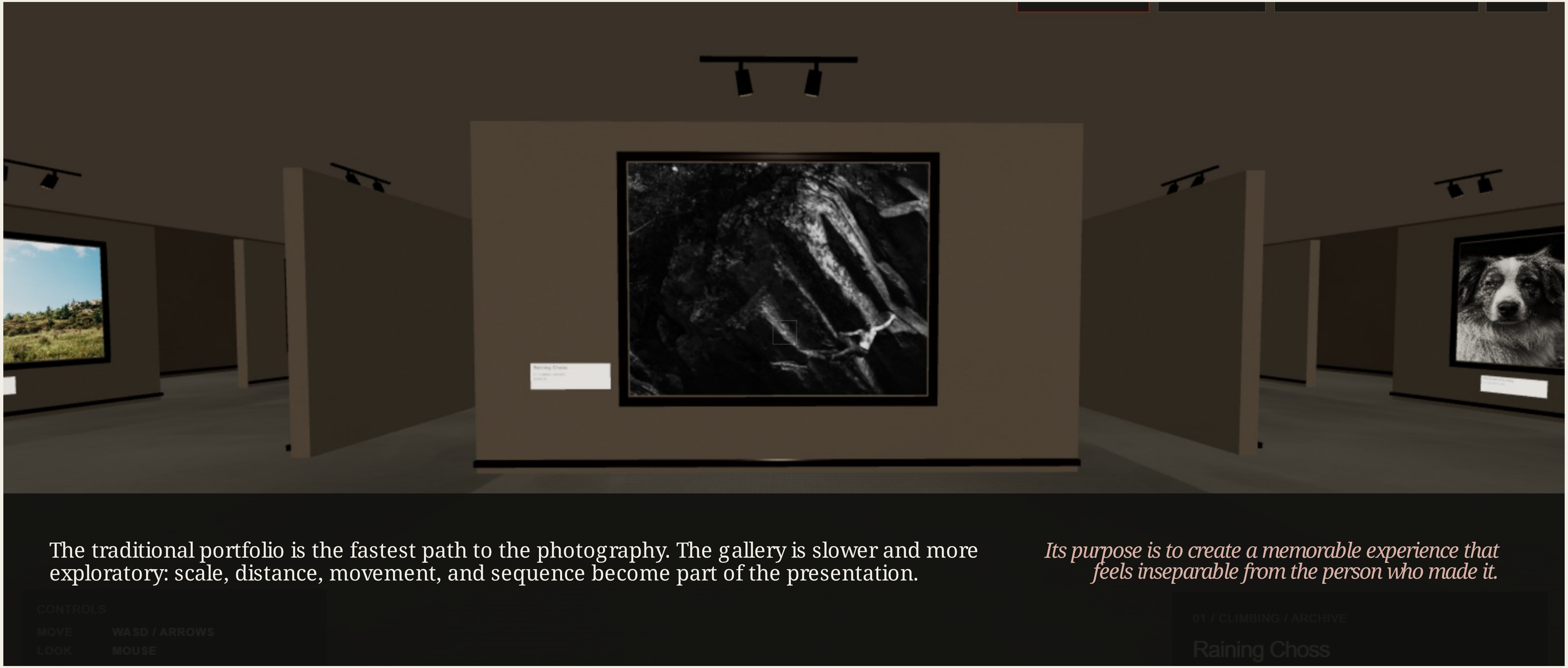
Position · size · layer · rotation · opacity

A specialized workspace can remain narrow without becoming generic design software.

08 / EXPERIENCE

WIP

Why the virtual gallery exists



The traditional portfolio is the fastest path to the photography. The gallery is slower and more exploratory: scale, distance, movement, and sequence become part of the presentation.

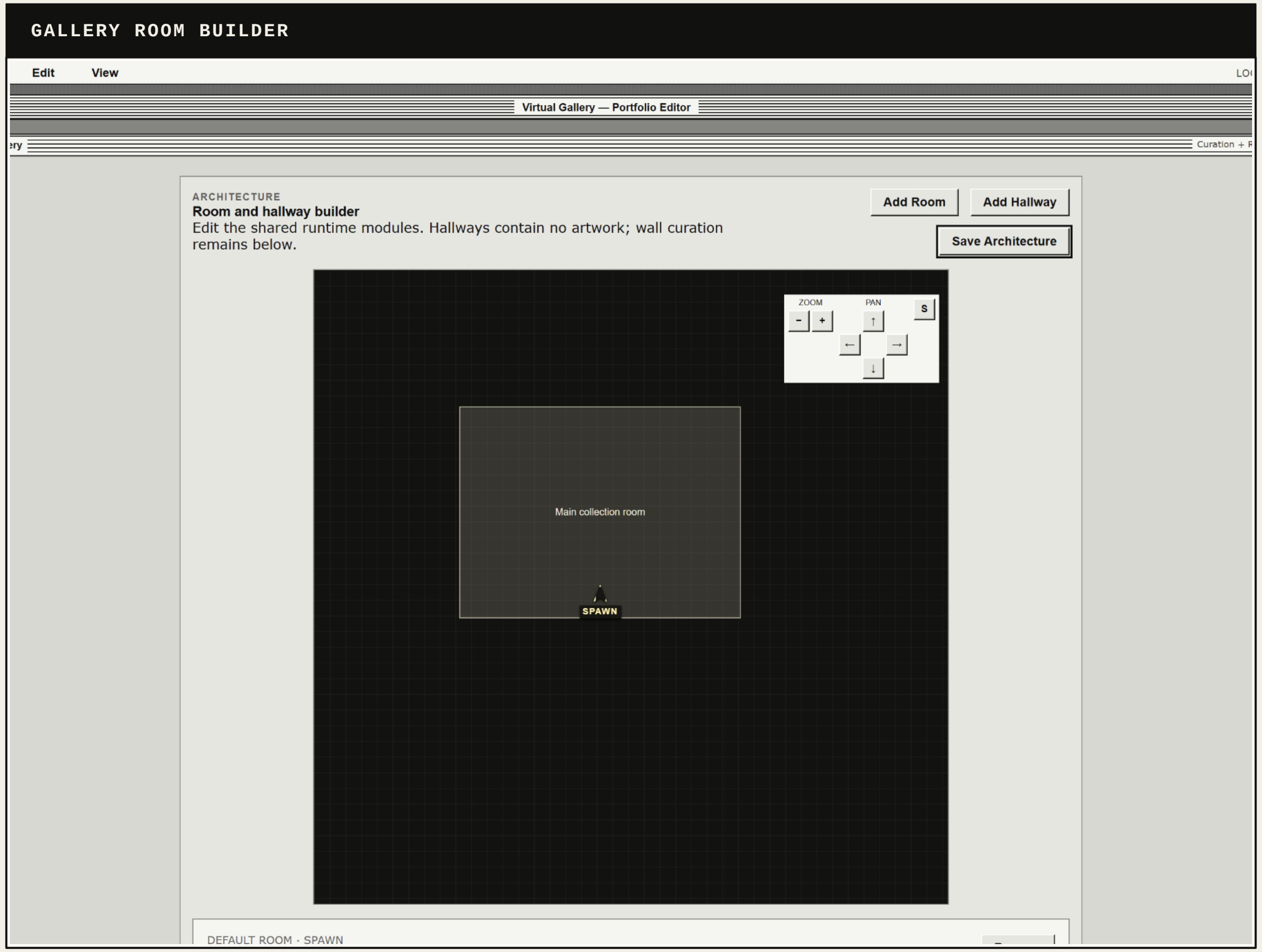
Its purpose is to create a memorable experience that feels inseparable from the person who made it.

Current gallery running in Auto / Low on integrated laptop graphics.

09 / ARCHITECTURE

WIP

Designed for expansion



Expansion was a core requirement. I did not want the first room to become a fixed environment that would need to be rebuilt when the portfolio grew.

Rooms, hallways, walls, connections, and artwork assignments are separated into editable records. The current gallery is small, but the architecture is intended to support larger exhibitions over time.

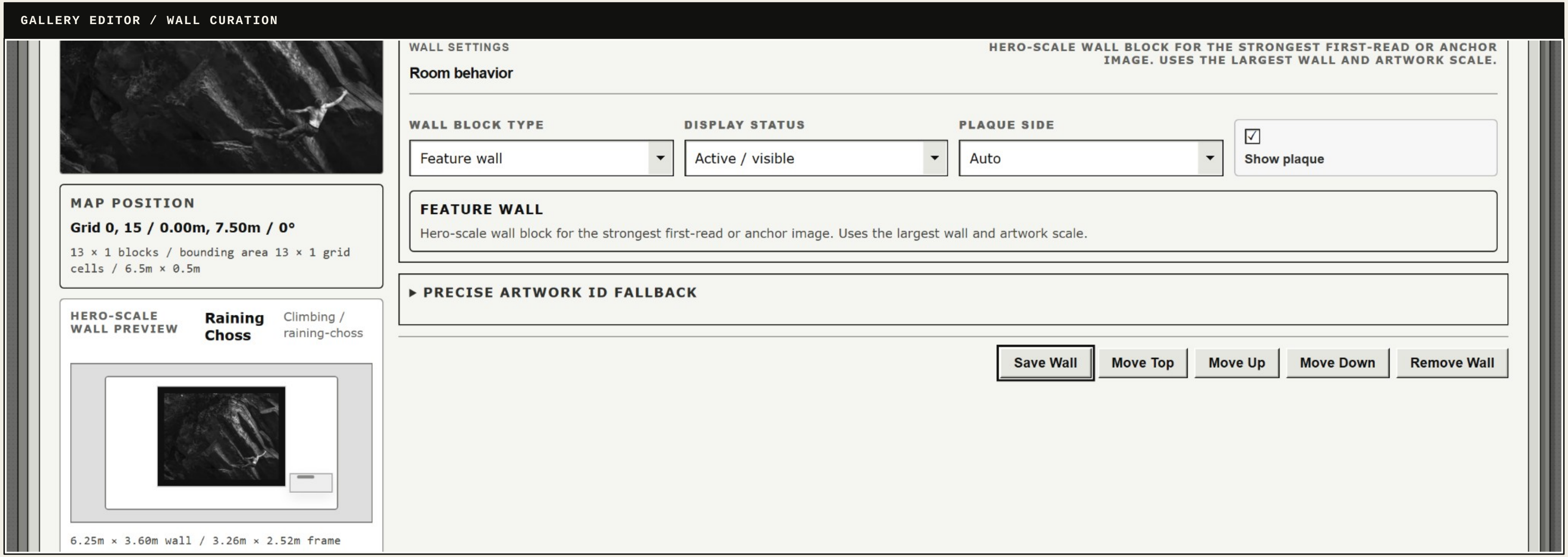
- local walls
- artwork assignments
- connection points

- local walls
- artwork assignments
- connection points

10 / CURATION

WIP

Specialized spatial curation



ARCHITECTURE DATA

Rooms · hallways · dimensions · connections · collision

CURATION DATA

Artwork · wall · position · scale · plaque · sequence

The workspace is an example of how a custom control panel can include a site-specific tool.

11 / PERFORMANCE

WIP

Adaptive quality preserves the experience

Testing on additional computers and mobile devices made the problem clear: a gallery that performs well on one machine can become slow or unstable on another. The system preserves the same room, artwork, movement, and navigation while reducing selected rendering costs.

CORE EXPERIENCE / PRESERVED

Room architecture
Artwork selection
Wall placement
Movement + navigation
Basic visibility

RENDERING COST / ADAPTED

Texture detail
Optional lighting
Shadows
Pixel ratio + renderer resolution
Upload timing

AUTO — measures + adapts

LOW — protects stability

MEDIUM — balanced detail

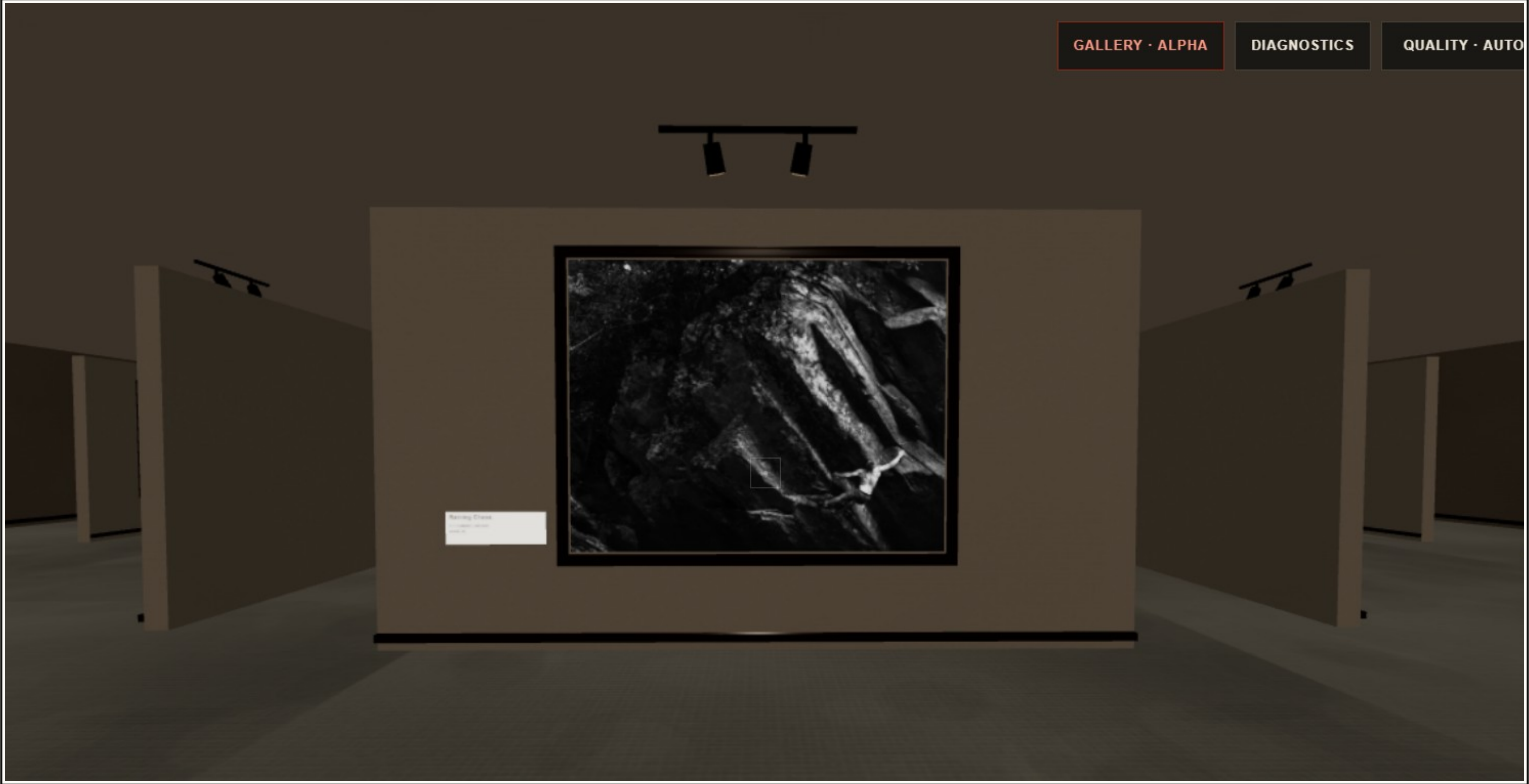
HIGH — full visual treatment

12 / EVIDENCE

CURRENT

Runtime evidence on constrained hardware

AUTO / LOW — VISITOR EXPERIENCE



LOCAL DIAGNOSTICS

My laptop remains at Low, but the system preserves the gallery rather than rejecting the device. The diagnostics distinguish the tier currently justified by readiness and runtime measurements from the maximum tier that remains theoretically available.

60 fps	cadence	**16.7 ms**	average frame	**1.8 ms**	gallery work
102	draw calls	**6,772**	triangles	**41**	textures

NO TELEMETRY

13 / DELIVERY

CURRENT

From network request to GPU resource

Browser-cache readiness and GPU readiness are separate states. Image requests can begin before gallery entry, while texture preparation remains controlled. On exit, artwork textures can be released from GPU memory while downloaded files remain cached for a faster return.

HTTP
Portfolio entry
→
BROWSER
Cache warming
→
SCENE
Gallery entry
→
GPU
Decode + upload
→
RENDER
Active textures
→
CLEANUP
Gallery exit
→
FASTER
Return visit

MEMORY
GPU artwork textures released

NETWORK
Browser cache retained

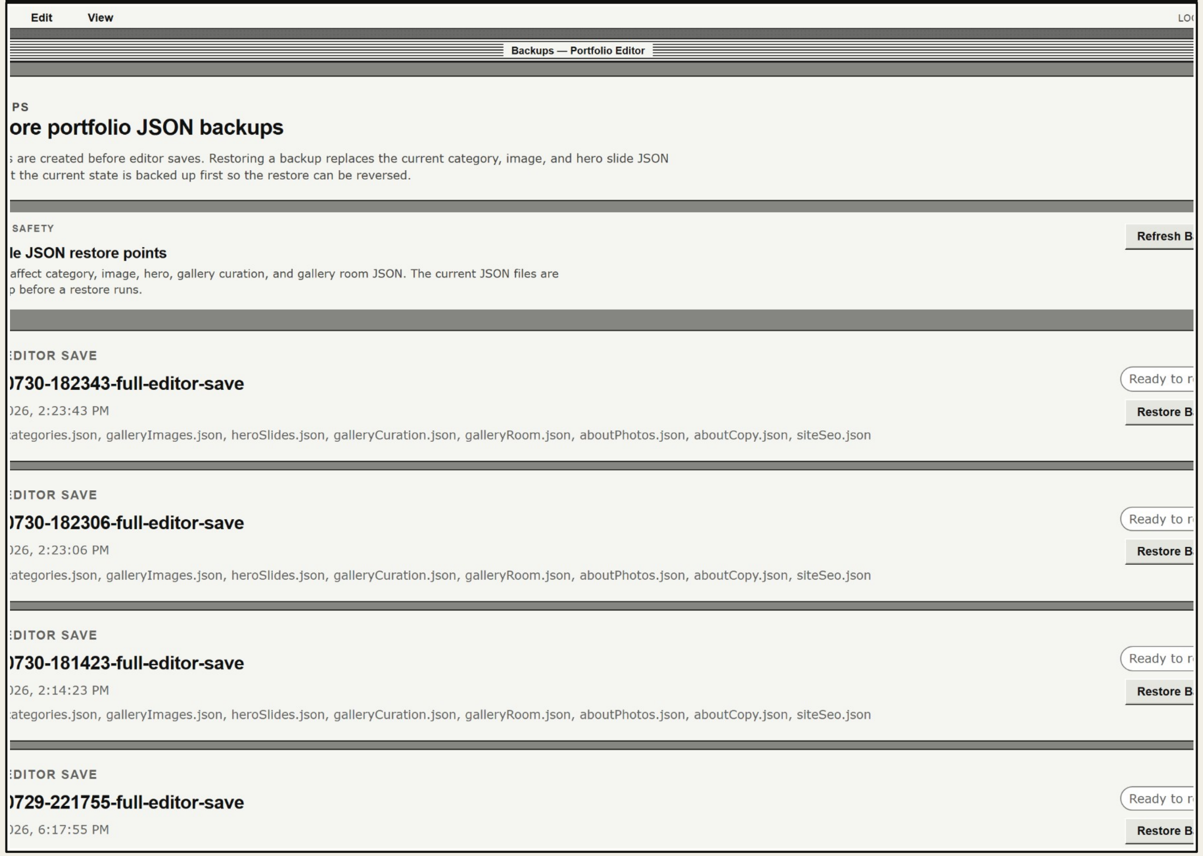
> *The gallery can return without paying the full network cost or permanently occupying graphics memory.*

14 / RELIABILITY

WIP

Recoverable editing, honest limitations

BACKUP + RESTORE



RELIABILITY

Because the editor changes the structured data used by the public site, recoverability is essential. Save operations create backups before replacing active records. Import actions validate data and preserve processed sources in an archive. Build and layout checks help catch invalid paths and inconsistent records before release.

AI + PRODUCT DIRECTION

I used AI primarily to assist with code, debugging, and interface asset creation. I directed the project from a product-management perspective: defining problems, setting requirements, prioritizing workflows, evaluating results, and deciding how the system should behave.

The interface remains visually inconsistent and has not received a professional product-design pass. The editor is also still local and portfolio-specific.

VALIDATE → BACK UP → SAVE → BUILD → REVIEW → RELEASE

15 / DIRECTION

PLANNED

From portfolio editor to custom-site control panel

The current editor was built for one website, but the product idea is broader: a control panel that can connect to developer-built websites without handing clients unrestricted control over the implementation.

SYSTEM

Developer-controlled

- Source code
- Templates
- Responsive rules
- Accessibility
- Application logic
- Infrastructure

→

PRODUCT

Custom-site control panel

Reusable structured tools
+
site-specific visual workspaces

→

CONTENT

Owner-controlled

- Media
- Copy
- SEO
- Navigation
- Approved presets
- Bounded layout data

> *I began by building a gallery for my own photographs. The editor emerged from everything required to keep that gallery - and the website around it - usable.*